

# ELI — Architecture

ELI v2.4.82

September 2026

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**Updated for v2.4.72 (September 2026).** Every CHAT turn runs the gradient orchestrator at a depth scaled to the reasoning mode, with the specialist bus composed inside it after shared retrieval (eli/memory/retrieval.py). Canonical S01-S12 tracing is in eli/kernel/pipeline\_trace.py; stage 12 learning runs through learning\_coordinator.py.

## Blueprint — ELI Architecture

The structural map of ELI v2. File paths, counts and names below were checked against the source tree on 2026-09-25; where something is only true at runtime it is marked (*runtime*).

ELI is a **local-first, offline-by-default, model-agnostic** cognitive runtime and assistant GUI, and also a self-hosted web app (api/server.py, described in ELI\_USER\_MANUAL.md). No cloud on the inference path, no hardcoded model. **196,249 lines across 438 Python files in eli/**, plus api/server.py (2,292 lines). 228 capabilities in capability\_manifest.json, 186 of them routable and 205 in the executor’s supported list.

## 0. Design principles

1. **Local-first, offline-by-default.** Outbound network is refused at the socket boundary (`eli/core/netguard.py`) unless the owner enables it; features that need the network opt in.
  2. **Model-agnostic.** No model name or size is hardcoded on the inference path. ELI loads whatever GGUF is configured and identifies its family from file metadata.
  3. **Grounded.** A dedicated subsystem (§9) exists to stop a local model stating things it cannot back up.
  4. **Persona is not scripted away.** ELI has a distinct voice; functional bugs are fixed without flattening it.
- 

## 1. Repository map

*Files and lines per package under eli/, measured 2026-09-25.*

| Package        | Lines  | Files | Role                                                                                      |
|----------------|--------|-------|-------------------------------------------------------------------------------------------|
| eli/runtime    | 35,375 | 95    | Grounding spine, evidence, response and introspection surfaces, daemons, self-improvement |
| eli/gui        | 27,447 | 27    | PySide6 desktop app, panels, tabs, startup and first-boot wizard, animated face           |
| eli/execution  | 26,293 | 13    | Router and executor: intent → action → side effects                                       |
| eli/cognition  | 20,824 | 47    | Agent bus, 12-stage orchestrator, inference, persona, reasoning modes, tone and emotion   |
| eli/kernel     | 17,545 | 8     | CognitiveEngine, pipeline tracing, scheduler, self-upgrade, world model                   |
| eli/core       | 11,222 | 30    | netguard, paths, settings, hardware profile, model download, full control                 |
| eli/perception | 10,220 | 23    | Vision, STT, TTS, wake word, voice tone, OS control, gaze                                 |
| eli/memory     | 8,770  | 11    | SQLite, FTS5, FAISS, knowledge graph, storage policy                                      |

| Package            | Lines          | Files      | Role                                                                  |
|--------------------|----------------|------------|-----------------------------------------------------------------------|
| eli/tools          | 7,573          | 29         | Image engine, news, capability registry, document tools               |
| eli/plugins        | 5,911          | 34         | Plugin manager, marketplace client, bundled plugins                   |
| eli/learning       | 4,251          | 14         | LoRA self-training pipeline                                           |
| eli/planning       | 4,000          | 19         | Proactive daemon, habit scheduler, job, proposal and attention queues |
| eli/integrations   | 3,339          | 18         | Ollama client, MCP client and server, cross-platform media, MPRIS     |
| eli/coding         | 2,104          | 12         | CodeAgent: plan → search → verify → repair                            |
| eli/setup          | 1,935          | 9          | Installer, hardware policy, first-run wizard                          |
| eli/utils          | 1,838          | 5          | Logging, platform compatibility                                       |
| eli/world          | 1,793          | 26         | World event bus, local world bridge, avatar and ontology              |
| eli/system         | 1,190          | 3          | Portable app control, process guard                                   |
| eli/contracts      | 787            | 3          | Typed runtime-status contracts                                        |
| eli/onboarding     | 709            | 2          | Onboarding interview                                                  |
| eli/cli, top level | 236            | 4          | Headless REPL, package entry                                          |
| <b>total</b>       | <b>193,503</b> | <b>434</b> |                                                                       |

The largest files: eli/kernel/engine.py (16,014 lines), eli/execution/executor\_enhanced.py (15,845), eli/gui/eli\_pro\_audio\_gui\_v2\_0.py (13,107), eli/execution/router\_enhanced.py (8,271), eli/gui/labs\_tab.py (5,744), eli/memory/memory.py (5,677), eli/cognition/agent\_bus.py (3,517), eli/runtime/deterministic\_grounding\_ga (3,367), eli/cognition/gguf\_inference.py (3,234).

## 2. Entry points and launch

| Command                | Target                           | Notes                 |
|------------------------|----------------------------------|-----------------------|
| eli, eli-v2.0, eli-cli | eli.gui.app:main                 | GUI (default)         |
| eli-mcp                | eli.integrations.mcp.server:main | ELI as an MCP server  |
| eli-jobs               | eli.planning.jobqueue_cli:main   | durable job queue CLI |

| Command                               | Target                        | Notes                                                  |
|---------------------------------------|-------------------------------|--------------------------------------------------------|
| <code>python -m eli</code>            | <code>eli/__main__.py</code>  | GUI, or headless with flags                            |
| <code>python -m eli --headless</code> | <code>eli.cli.headless</code> | terminal REPL, no Qt                                   |
| <code>./eli.sh</code>                 | <code>python -m eli</code>    | venv launcher                                          |
| <code>bash install.sh</code>          |                               | venv, dependencies, clean config<br>seed, verification |

Boot path (GUI): `eli/gui/app.py` → first-boot wizard when no model is configured (`eli/gui/panels/startup.py`) → `eli_pro_audio_gui_v2_0.py:main()` builds `EliMainWindow` → constructs the `CognitiveEngine` singleton → loads the GGUF named in `config/settings.json` → starts the daemons.

### 3. The request lifecycle

Every request goes through `CognitiveEngine.process()` (`eli/kernel/engine.py`).

```

user input
|
▼
[Router] eli/execution/router_enhanced.py :: route()
|   priority pipeline → {action, args, confidence, meta.matched_by}
|   ▼
CognitiveEngine.process(action, args, ...)
|
├─ deterministic fast path (OS, media, status, job actions)
|   └─ execute_action() → returned verbatim, no model
|
├─ CHAT (any reasoning mode, Quick to Expert)
|   └─ AgentOrchestrator.run() at mode depth
|       S05 planner → shared retrieval → dispatch_specialists()
|       └─ context assembly → broker.infer() → output governor
|
├─ non-CHAT action
|   └─ AgentOrchestrator → bus + ReAct tool loop
|
└─ orchestrator returns None or raises → AgentBus.dispatch() fallback

```

The canonical stage names (`pipeline_trace.STAGE_NAMES`) are S01 PERCEIVE\_INGEST, S02 INPUT\_GUARDS, S03 ROUTER, S04 GROUNDING\_GATE, S05 PLANNER, S06 AGENT\_BUS, S07 CONTEXT\_ASSEMBLY, S08 INFERENCE\_BROKER, S09 REASONING\_SYNTHESIS, S10 OUTPUT\_GOVERNOR, S11 RESPONSE\_DELIVERY, S12 LEARNING\_STATE\_UPDATE. Enable tracing with `ELI_PIPELINE_TRACE=1` or `scripts/eli_startup.sh --trace`.

### 4. Routing layer (eli/execution)

- **router\_enhanced.py**: `route(text) -> {action, args, confidence, meta}`. A regex-first priority pipeline of ordered prepasses (realtime web lookup, media, file and PDF, memory, control), then a core router, with an LLM intent fallback (`cognition/llm_intent.py`).
- **execution\_planner.py**: typed `ExecutionPlan` and `RouteDecision` artifacts.
- **route\_contracts.py**, **operator\_policy.py**, **shell\_gate.py**: what may be routed and executed.
- **portable\_intent\_contract.py**, **media\_runtime.py**, **operator\_actions.py**: media and OS intent wiring. Plugin actions are handled directly in `executor_enhanced.py`.

Routes that were once bugs are pinned by tests and by tools/eval/cases.yaml: media controls need whole-word, command-shaped input; a bare “do a search” needs a subject; meta-questions stay in CHAT; realtime facts go to WEB\_SEARCH; code words such as “repo” match whole words only, so “morning report” is not a codebase audit.

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## 5-6. Agent bus and agents (eli/cognition/agent\_bus.py)

AgentBus.dispatch(user\_input, intent, ...) -> DispatchResult selects an agent set (\_select\_agents\_for\_intent, or a broad fan-out), runs it over a dependency DAG (flat parallel if the DAG fails) and aggregates. DispatchResult carries agent\_results, action\_result, memory\_context, aggregated\_confidence, grounding\_confidence (agent evidence only), agents\_used, confidence\_label and orchestrator\_plan.

**15 registered agents** (\_ALL\_AGENTS):

| Agent                 | Role                                                        |
|-----------------------|-------------------------------------------------------------|
| BusMemoryAgent        | recall from SQLite, FTS and FAISS into context              |
| SystemAgent           | runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, ...)      |
| HabitAgent            | learned habits                                              |
| SelfImprovementAgent  | recent failures, improvement proposals, corrections         |
| ProactiveAgent        | proactive patterns and signals                              |
| FrontierAgent         | frontier reasoning hooks                                    |
| PluginAgent           | dispatches to registered plugins                            |
| CapabilityAgent       | what ELI can do (capability manifest)                       |
| VoiceAgent            | STT and TTS engine state                                    |
| OrchestratorAgent     | plans and coordinates grounded synthesis                    |
| FileCodeAgent         | searches the codebase for code-grounded answers             |
| ReflectionAgent       | session reflection                                          |
| IntrospectionBusAgent | live runtime and self introspection                         |
| KnowledgeGraphAgent   | entities and relations from the knowledge graph             |
| CriticAgent           | second-tier verifier; runs after its retriever dependencies |

SpecAgent runs user-defined agent specifications against the local model. The coding pipeline, CodeAgent (eli/coding/agent.py, plan → search → verify → repair), is separate from the bus and is invoked for CODE\_SOLVE and GENERATE\_SCRIPT.

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## 7. The orchestrator (eli/cognition/orchestrator.py)

Runs for all CHAT modes at gradient depth, and for non-CHAT actions. Shared retrieval lives in eli/memory/retrieval.py (retrieve\_for\_turn()), called by both BusMemoryAgent and the orchestrator so a turn is not searched twice:

- **Turn cache:** 8 seconds per process. Its key includes every retrieval limit, so a deeper pass with larger limits is never served the shallower result.
- **Time window first:** when the question names a period (“last week”, “the past two weeks”), query\_planner.parse\_window produces a window. Memories inside it are fetched by date (Memory.memories\_between) before any ranking, then ranked by topic. The search reports what the window did (memory\_diag).
- **Sequential by design:** the llama.cpp embedder is not thread-safe.

- **Heuristic rerank** (cognition/reranker.py): lexical overlap  $\times$  recency  $\times$  importance. A neural cross-encoder is not in use.
- **FAISS tombstones**: `vector_store.mark_memory_deleted()` marks stale vectors without a rebuild; `compact_tombstones()` reclaims space.

Mode scaling (reasoning\_modes.py): Quick uses the fast planner and a lean specialist profile; Normal, Advanced, Research and Expert use progressively deeper planner budgets and wider `mode_chat_agent_profile()` fan-out. Quick still runs the orchestrator.

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## 8. Inference layer (eli/cognition)

- **inference\_broker.py**: the canonical inference path, `broker.infer()`.
  - **gguf\_inference.py**: the llama-cpp wrapper: live runtime parameters, `chat_completion()`, runtime snapshot publishing.
  - **model\_identity.py, model\_output\_tokens.py**: model family and chat template are read from GGUF metadata, not from the file name; special tokens are stripped per family.
  - **reasoning\_modes.py**: quick, chain\_of\_thought, self\_consistency, tree\_of\_thoughts, constitutional\_ai (private strategies; the raw reasoning is not shown).
  - **context\_synthesiser.py, user\_info\_builder.py**: prompt and context assembly.
  - **context\_budget.py**: how much of the window memory may use. Output reserve is an eighth of the window; on a recall question the memory floor is a third of it.
  - **llm\_intent.py**: LLM fallback for ambiguous routing. Decoding is constrained by a GBNF grammar built from the live action catalogue, so the model can only name a real capability and emit well-formed JSON. Backends without grammar support fall back to free text.
  - Budgeting lives in `engine.py::_build_enhanced_system` and `core/dynamic_runtime_budget.py`.
- 

## 9. Grounding and anti-confabulation (eli/runtime, eli/core/grounding.py)

- **deterministic\_grounding\_gate.py**: decides when an answer must be backed by evidence and renders self-reports and audits from live measurements. `render_action` is one explicit ordered pipeline of layers (`_stack`), not captured copies.
- **grounding\_escalation.py**: a checkable factual turn with low grounding escalates (external fact  $\rightarrow$  web agent, self or project fact  $\rightarrow$  broad local fan-out) and hedges honestly if nothing grounds it. Env: `ELI_GROUNDING_ESCALATION`.
- **evidence\_ledger.py, evidence\_arbitration.py**: what counts as evidence, where it is recorded (the ledger also records every executed action), and conflict resolution.
- **persistence\_gate.py**: keeps internal report dumps out of memory.
- **response\_contracts.py, response\_policy.py, final\_response\_provider.py, user\_visible\_response\_surface.py**: shape and clean the user-visible answer.
- **truth\_report.py, eli\_identity\_audit.py, control\_contracts.py**: runtime truth, identity grounding, control-action contracts.
- **cognition/output\_governor.py, response\_governance.py, response\_sanitizer.py, persona\_hygiene.py**: the final governance pass, including removal of false diagnoses when retrieval demonstrably ran.

The known weak seam is internal state leaking into spoken output on the plain CHAT path (identity confabulation, telemetry dumps). It is defended, not closed.

---

## 10. Execution layer (eli/execution/executor\_enhanced.py)

- `execute(action, args) -> dict`. The executor's supported list holds **206 actions**; the manifest declares **228 capabilities**, of which **186 are routable** and 206 are in the supported list.

- **Fast path** (engine.py): deterministic OS, media, status and job actions (VOLUME, MEDIA\_CONTROL, NEXT\_MEDIA, OPEN\_APP, DATE, SHELL\_EXEC, ANALYZE\_IMAGE, CHECK\_JOB, BACKGROUND\_JOBS, ...) return the executor result verbatim.
- Action families: media and OS control, files and documents (SUMMARIZE\_FILE, ANALYZE\_PDF[\_FOLDER] including per-file mode and saving into memory, CREATE\_DOCUMENT), web (WEB\_SEARCH), news, weather, memory (MEMORY\_STORE, MEMORY\_RECALL), code (CODE\_SOLVE, GENERATE\_SCRIPT), background jobs, self and runtime reports (RUNTIME\_STATUS, SELF\_REPORT, RUNTIME\_AUDIT, EXPLAIN\_\*), vision, image generation, scheduling.
- **Background jobs** (runtime/background\_tasks.py): submit/get/list. Heavy PDF-folder analysis backgrounds itself; CHECK\_JOB returns the real result.
- **YouTube playback** lives in eli/integrations/media/youtube\_playback.py; the executor delegates to it.

## Plugins (eli/plugins)

Bundled: calendar, document\_reader, media, notes, pomodoro, system\_stats, tts, weather, web, web\_automation. State is in config/plugins\_state.json (gitignored). Custom agents and plugins are trust-gated (\$ plugin table below).

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## 11. Memory subsystem (eli/memory)

- **memory.py** (Memory, get\_memory()): long-term store over SQLite and FTS5, plus failure, observation and habit logging.
- **policy.py**: the storage policy. Every row carries an origin (user\_said, eli\_said, telemetry, news, tool); repeats are merged (seen\_count); a forgetting curve scales weight by importance, origin and reinforcement; faded derived rows move to memories\_archive. A daily run\_upkeep() does the merging, decay, archiving, identity merge and learning-table tidy. Full detail in memory.md.
- **vector\_store.py**: FAISS index at artifacts/vectors/index.faiss; embedder models/embeddings/nomic-embed-text-v1.5.Q4\_K\_M.gguf (*runtime*).
- **Knowledge graph**: kg\_entities and kg\_relations (FTS5-backed) in the user database.
- **Working memory** (cognition/working\_memory.py): session-pinned facts with dates preserved, re-stored across sessions.
- **Personal memory**: runtime/personal\_memory\_\*, persona\_updater.

**Four SQLite files** under artifacts/db/ (*runtime*): user.sqlite3 (memory, conversation, knowledge graph, learning tables), agent.sqlite3 (self-improvement records and agent metrics), system\_index.sqlite3 (OS index) and coding\_memory.sqlite3 (coding bug fixes). The blank user.sqlite3 template holds 27 tables plus three FTS5 indexes: memories, memories\_archive, memory\_meta, semantic, conversation\_turns, conversations, session\_summaries, kg\_entities, kg\_relations, recall\_log, runtime\_events, learning\_replay, observations, habits, habit\_events, habit\_rules, user\_patterns, user\_model, eli\_stances, belief\_revisions, corrections, failures, error\_tracking, improvements, capability\_proposals, news\_articles, news\_reflections. Self-improvement records (improvements, failures, corrections, capability\_proposals, error\_tracking) are written to agent.sqlite3; the copies in the user database are empty. Recall is hybrid: keyword, FTS5, FAISS, knowledge graph and recent conversation, merged and reranked (§7).

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## 12. Perception (eli/perception)

- **Vision**: vision.py (model-agnostic vision-language, Moondream and Qwen2.5-VL hot-swap; the multimodal encoder is forced to CPU to avoid a CUDA clip segfault), analyze\_image.py, ambient\_vision.py, screen\_locator.py, gaze\_engine.py, analyze\_csv.py, analyze\_pdfs.py.

- **STT**: `local_whisper_stt.py` (faster-whisper, offline when the Net switch is off), `audio_stt.py`, `voice_worker_streaming.py`. Default wake phrases: “computer”, “hey computer”, “eli”, “hey eli” (`wakeword.py`); a music-bleed filter and an echo gate sit in front of the transcript.
  - **TTS**: `tts_router.py` (Piper voices), `tts_xtts.py` (optional neural voices).
  - **OS**: `os_controller.py` (app, window, keyboard, mouse), `ui_tree.py`, `desktop_capabilities.py`.
- 

### 13. Persona (`eli/cognition/persona*`)

- `persona.txt` (authored) and `persona.auto.txt` (overlay) hold the voice.
  - `persona_updater.py` updates the overlay, knowledge graph and user profile each turn.
  - `persona.py`, `persona_values.py`, `persona_status.py`, `persona_hygiene.py`.
  - `tone_analyzer.py` and `tone_adaptor.py` read the user’s tone; `emotion_timeline.py` keeps a history that lets ELI notice a sustained mood.
  - Generation injects a budget-trimmed persona and situation brief (`engine.py::_build_enhanced_system`); a live-runtime fact is injected for model and identity questions so ELI reports the loaded model.
- 

### 14. Daemons and background loops

- **Proactive daemon** (`planning/proactive_daemon.py`): pattern signals (time habit, topic focus, recurring errors, active project); runs an autonomy tick every 30 minutes.
  - **Self-improvement loop** (`runtime/self_improvement.py`): started at boot, analyses failures immediately and then every 24 hours. Repair by the coding agent is opt-in and bounded. A failure is stored with a capsule (input, versions, model, error class), and a candidate patch is tried on a hard-linked copy of the tree and compared with the same tests on the untouched tree: only a candidate that fixes something and breaks nothing is applied, every candidate stays in the archive (`code_patches`) with its verdict and cost, and the tests, the evaluator and the safety gates cannot be patched by a candidate. `ELI_SELFPATCH_UNPROVEN=1` restores applying without proof.
  - **Habit scheduler** (`planning/habits_scheduler.py`, `habits.py`).
  - **Scheduler** (`kernel/scheduler.py`); **background tasks** (`runtime/background_tasks.py`).
  - **Code monitor** (`runtime/code_monitor.py`); **ambient vision loop**.
  - **World event bus** (`world/world_event_bus.py`).
  - **Memory upkeep**: once a day, started asynchronously at engine init and after responses.
- 

### 15. Learning and self-training (`eli/learning`)

A LoRA fine-tuning pipeline: `bootstrap_phi3_base.py`, `base_model_resolver.py`, `dataset_builder.py`, `dataset_filters.py`, `merge_reviewed_datasets.py`, `export_trainable_dataset.py`, `training_preflight.py`, `lora_trainer.py`, `lora_trainer_guard.py`, `lora_eval.py`, plus `target_registry.py` (operator-declared targets, any model family) and `review_queue.py` (the human review gate).

---

### 16. EliWorld (`eli/world`, the **EliWorld tab**, `kernel/world_model.py`)

An internal world the agent inhabits. `world_event_bus.py` receives runtime events; `local_world_bridge.py` bridges to the model; snapshots, ledger and journal are under `artifacts/world/`. Experimental.

---

## 17. Core infrastructure (eli/core)

- **netguard.py**: offline by default: guarded\_urlopen, a process-wide socket guard (fail-closed on non-loopback while offline), and allow\_network() for deliberate user-initiated fetches.
  - **paths.py, portable\_paths.py, legacy\_paths.py, db\_paths.py**: path resolution (dev tree vs platform directories, ELI\_\* overrides, read-only frozen bundles).
  - **runtime\_settings.py, config.py**: settings load, merge and heal; DEFAULTS are clean (offline, no model, wizard on). config/settings.json is per user and gitignored, seeded from config/templates/settings.template.json.
  - **hardware\_profile.py, startup\_hardware\_optimizer.py, dynamic\_runtime\_budget.py**: detect GPU and VRAM and pick context, GPU layers and batch (one fit calculation shared by the loader and the startup dialog).
  - **model\_download.py**: curated GGUF downloader (7 catalogue entries, resumable, magic and size validated, netguard-gated). Install-time menu only.
  - **grounding.py**: shared grounding helpers.
- 

## 18. GUI (eli/gui)

PySide6 (LGPL). eli\_pro\_audio\_gui\_v2\_0.py (main window, 13.1k lines), app.py (boot), labs\_tab.py (Labs, 5.7k lines), panels/ (startup.py: StartupModelSelectionDialog, FirstBootWizard and the hardware tuning dock; settings.py, agent\_wizard.py, permission\_dialog.py), tabs/ (eli\_world\_tab.py, experimental\_tab.py, marketplace\_tab.py, tasks\_tab.py, training\_tab.py), docks/ (operator console, proactive dock), widgets/ (animated face, voice downloader, Ollama selector). qt\_compat.py allows a PyQt fallback from source.

---

## 19. On-disk layout (artifacts/, runtime)

```
artifacts/  
├─ db/{user,agent,system_index,coding_memory}.sqlite3  
├─ vectors/index.faiss           # semantic index  
├─ conversations/               # conversation exports  
├─ documents/ , scripts/        # generated artifacts  
├─ runtime/                     # runtime state and capability inventory  
├─ runtime_snapshot.json        # live model/runtime truth  
├─ world/                       # snapshots, ledger, journal  
└─ analyze_image_*/             # vision outputs  
config/  settings.json (gitignored) · templates/ · plugins_state.json · voices/ ...  
models/  <your>.gguf · embeddings/ · whisper/ · image/  
tts_piper/ Piper voice files shipped with the repo
```

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## 20. Known seams and weak points

1. **Large files.** engine.py, executor\_enhanced.py, the main GUI file and the router hold about a third of the code. The executor is an action ladder with a high regression surface.
2. **Internal state → spoken output.** The recurring failure class. The grounding gate and escalation defend it; they do not close it on the plain CHAT path.
3. **Swallowed errors.** 162 handlers are a bare except: pass and 4,392 catch Exception, most of them logging at debug level. A ratchet test (tests/claims/test\_no\_silent\_swallow.py) stops the silent count rising.
4. **Latency versus model size.** On an 8 GB GPU a 24B model at Q5 offloads few layers and answers in minutes; a 7-9B Q4 is the sweet spot. The startup dialog shows the real fit.

5. **Two GUI monoliths.** `eli/gui/eli_pro_audio_gui_v2_0.py` and `eli/gui/labs_tab.py` are single files of 13.1k and 5.7k lines.

## 21. Where to make a change

| To change...                                | Edit...                                                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| how input is routed to an action            | <code>eli/execution/router_enhanced.py</code>                                                                            |
| what an action does                         | <code>eli/execution/executor_enhanced.py</code>                                                                          |
| which agents run and how grounded           | <code>eli/cognition/agent_bus.py</code>                                                                                  |
| the 12-stage orchestration                  | <code>eli/cognition/orchestrator.py</code>                                                                               |
| prompt and persona budget, the engine spine | <code>eli/kernel/engine.py</code>                                                                                        |
| anti-confabulation                          | <code>eli/runtime/grounding_escalation.py</code> ,<br><code>deterministic_grounding_gate.py</code>                       |
| final answer shaping                        | <code>eli/runtime/*response*</code> ,<br><code>cognition/output_governor.py</code>                                       |
| memory store and recall                     | <code>eli/memory/memory.py</code> , <code>policy.py</code> , <code>retrieval.py</code> ,<br><code>vector_store.py</code> |
| model loading and inference                 | <code>eli/cognition/gguf_inference.py</code> ,<br><code>inference_broker.py</code>                                       |
| offline and network policy                  | <code>eli/core/netguard.py</code>                                                                                        |
| paths, settings, model download             | <code>eli/core/{paths, runtime_settings, model_download}.py</code>                                                       |
| the GUI                                     | <code>eli/gui/eli_pro_audio_gui_v2_0.py</code> ,<br><code>eli/gui/panels/</code> , <code>eli/gui/tabs/</code>            |
| evaluation and regression                   | <code>tools/eval/</code>                                                                                                 |

## Plugins, agent specifications and training surfaces

**`eli/plugins/`: a loader plus a gated marketplace client**

| Module                        | Lines | Role                                                              |
|-------------------------------|-------|-------------------------------------------------------------------|
| <code>manager.py</code>       | 708   | discovery, enable and disable, install and uninstall              |
| <code>permissions.py</code>   | 395   | capability vocabulary, consent decisions, grants, audit ledger    |
| <code>manifest.py</code>      | 302   | manifest schema and static capability verification against source |
| <code>integrity.py</code>     | 291   | sha256 pinning, ed25519 signatures, operator-trusted publishers   |
| <code>security_scan.py</code> | 547   | multi-engine malware scanner                                      |
| <code>marketplace.py</code>   | 793   | federated registries, preview and install, licence keys           |
| <code>mcp.py</code>           | 481   | MCP server config, runtime preflight, handshake verification      |

`_plugins_dir()` resolves through `paths.plugins_dir()`, and `_plugin_search_dirs()` scans bundled built-ins and user installs, so a downloaded plugin is never written into a read-only installation.

### **eli/cognition/: agents get a specification**

| Module                      | Lines | Role                                                               |
|-----------------------------|-------|--------------------------------------------------------------------|
| <code>agent_spec.py</code>  | 436   | AgentSpec: objective, prompt, triggers, success criteria, examples |
| <code>agent_trust.py</code> | 262   | path-keyed trust with provenance, scanning, revocation             |

`agent_bus.SpecAgent` runs a spec against the local model and scores the output against the spec's own criteria. Spec agents execute no arbitrary code and bypass the trust chain.

### **GUI surfaces**

`tabs/training_tab.py` (Labs ► Training), `tabs/marketplace_tab.py` (Settings ► Marketplace), `panels/permission_dialog.py` (consent dialog and worker-to-GUI-thread bridge).

### **Paths**

`paths.learning_dir()` joins `models_dir()`, `voices_dir()` and `plugins_dir()`. The rule: runtime state never lives in the installation, because `project_root()` is a read-only mount in a packaged build.